Project Submission & Grading Guide

Knowledge Graph Construction · Alignment · Reasoning & KGE · RAG

This project integrates the four lab sessions into one full pipeline.
Your submission includes **(1) a GitHub repository** and **(2) a short final report**.
Total: 20 points (+1 bonus possible)

---

1. Required Submission Components

A. GitHub Repository (mandatory)

1. Code

- Crawler + cleaning + NER
- RDF graph + ontology + alignment scripts
- SPARQL expansion
- SWRL reasoning (OWLReady2)
- KGE prep + training + evaluation
- RAG pipeline (NL→SPARQL, self-repair, CLI/UI)

2. Knowledge Graph Files

- Initial RDF graph
- Ontology file
- Alignment file
- Expanded KB (NT/Turtle)
- KB statistics

3. KGE Datasets

- `train.txt`, `valid.txt`, `test.txt`

4. README.md (required)

Must include:

- Installation
- How to run each module
- How to run the RAG demo
- Hardware requirements
- At least one screenshot

5. Environment / Reproducibility

- `requirements.txt` or `environment.yml`

- Instructions for running Ollama
- If data is large: provide sample + download link

6. GitHub Good Practices

- Clear repository structure
- Clean commit history
- `.gitignore`
- (Optional) CI or code formatting

Here is an example of your project directory structure:

```
project-root/
├─ src/
│  ├─ crawl/
│  ├─ ie/
│  ├─ kg/
│  ├─ reason/
│  ├─ kge/
│  └─ rag/
├─ data/
│  ├─ samples/
│  └─ README.md
├─ kg_artifacts/
│  ├─ ontology.ttl
│  ├─ expanded.nt
│  └─ alignment.ttl
├─ reports/
│  └─ final_report.pdf
├─ notebooks/
├─ README.md
├─ requirements.txt
├─ .gitignore
└─ LICENSE
```

---

2. Final Report (6–10 pages)

Your report must include:

1. Data Acquisition & IE

- Domain + seed URLs
- Crawler design + ethics
- Cleaning pipeline
- NER examples
- 3 ambiguity cases

2. KB Construction & Alignment

- RDF modeling choices
- Entity linking with confidence
- Predicate alignment
- Expansion strategy
- Final KB statistics

3. Reasoning (SWRL)

- SWRL rule on `family.owl` + output
- One SWRL rule on your KB

4. Knowledge Graph Embeddings

- Data cleaning + splits
- Two models minimum
- Metrics: MRR, Hits@1/3/10
- Size-sensitivity (20k / 50k / full)
- t-SNE or nearest-neighbor examples

5. RAG over RDF/SPARQL

- Schema summary
- NL→SPARQL prompt template
- Self-repair mechanism
- Evaluation (≥5 questions: baseline vs RAG)
- Screenshot of demo

6. Critical Reflection

- KB quality impact
- Noise issues
- Rule-based vs embedding-based reasoning
- What you would improve

---

3. Grading Breakdown (20 pts)

A. Data Acquisition & IE — 3 pts

- Crawler quality + ethics: 1
- Cleaning + NER: 1
- Ambiguity & reflection: 1

B. KB Modeling, Alignment & Expansion — 5 pts

- RDF/ontology design: 1.5
- Entity/predicate alignment: 1.5
- SPARQL expansion strategy: 1.5

- KB statistics: 0.5

C. Reasoning & KGE — 4 pts

- SWRL rules + reasoning: 1
- ≥ 2 KGE models: 1
- Metrics + analysis: 2

D. RAG (NL→SPARQL + self-repair) — 4 pts

- Schema summary + prompting: 1
- Query execution & self-repair: 1
- Baseline vs RAG evaluation: 1
- Demo clarity: 1

E. GitHub Quality & Reproducibility — 2 pts

- README + instructions: 1
- Environment setup: 0.5
- Repo organization & commits: 0.5

F. Final Report — 2 pts

- Clarity, structure, visuals: 1
- Scientific depth & reflection: 1

Bonus (+1 pts)

For:

- Excellent demo/UI
- Extra models or analysis
- Advanced visualization
- Replicable pipeline end-to-end

---

4. Submission Procedure

1. Push all content to GitHub.
2. Create a **final release**: `v1.0-final`.
3. Submit on DVL:
    - **GitHub link to the release**
    - **Final report PDF**